
Abstract

Characterizing the global flow state in the DRESHDYN (DREsden Sodium Facility for DYnamo and Thermohydraulic Studies) precession experiment is essential for understanding the underlying flow dynamics, validating numerical simulations, and assessing the conditions for magnetic field generation. In this work, we outline the concept of a distributed Ultrasound Doppler Velocimetry measurement system for capturing large-scale flow structures in the DRESHDYN precession experiment. Based on this concept, we derive requirements for the ultrasound subsystem of the sensor nodes and present a tailored hardware design. The functionality of the developed subsystem is validated experimentally in a water-channel setup by comparison with Particle Image Velocimetry (PIV), showing a systematic deviation of 2.2 % across the investigated operating range. In addition, we demonstrate first-time capturing of radial velocity profiles in a down-scaled model of the precession experiment. The presented results establish the sensor-node architecture as a foundational building block for future distributed flow measurements in the DRESHDYN experiment.

Keywords: Distributed sensor system, Embedded sensing, Ultrasonic Doppler Velocimetry, Precession-driven flows, Liquid metal flows

1. Introduction

Cosmic magnetic fields play a central role in the formation of astronomical structures [1]. The dynamo effect is discussed as a source of energy for these fields. This refers to the self-excitation of a magnetic field from the interactions of the currents of an electrically conductive fluid. The basic principle of the geodynamo is well understood [2], yet unresolved questions remain concerning the energy budget available for magnetic field generation, particularly for the early geodynamo prior to the formation of Earth's solid inner core. Therefore, Earth's precession has been proposed as an additional driver of fluid motion in the liquid outer core to help overcome the limitations encountered in models that rely solely on convection-driven flows [3].

In an experiment unique in its scale, precession-driven dynamos are to be investigated within DRESHDYN (DREsden Sodium Facility for DYnamo and Thermohydraulic Studies at Helmholtz-Zentrum Dresden-Rossendorf (HZDR)) [4]. Within this experiment a vessel filled with 8 t of liquid sodium will be rotated at a frequency of up to $f_c = 10$ Hz and precessed at a frequency of up to $f_p = 1$ Hz. The resulting flow conditions will be recorded and their suit-

ability for magnetic field excitation will be investigated. The experimental machine is depicted in Fig. 1.

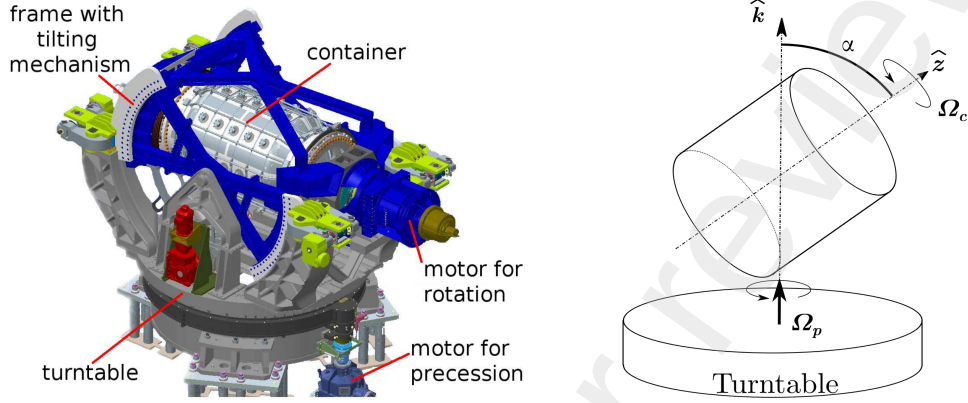


Figure 1: Experimental setup of the DRESHDYN precession experiment with annotated rotation axis $\hat{z}$ and rate $\Omega_c = 2\pi f_c$, precession axis $\hat{k}$ and rate $\Omega_p = 2\pi f_p$, nutation angle α and liquid sodium container [5].

Numerical simulations and preliminary investigations using a down-scaled experiment with water were used to examine the flow behavior as a function of the Reynolds number $Re = \Omega_c R^2 / \nu$ and the precession ratio $Po = \Omega_p / \Omega_c$ [6, 7, 8]. It was shown that the flow in a precessing cylinder can be described primarily by the three inertial modes shown in Fig. 2: An axisymmetric, geostrophic flow (left) composed by the rotational motion driven by the cylindrical cavity and the zonal flow resulting from nonlinear interactions of inertial waves, a standing wave (center) forced by precession and an axisymmetric double roll (right) [9, 10]. In [11] it was shown that the axially symmetric double-roll mode is conducive to the dynamo effect and, under certain conditions, can result in magnetic field excitation.

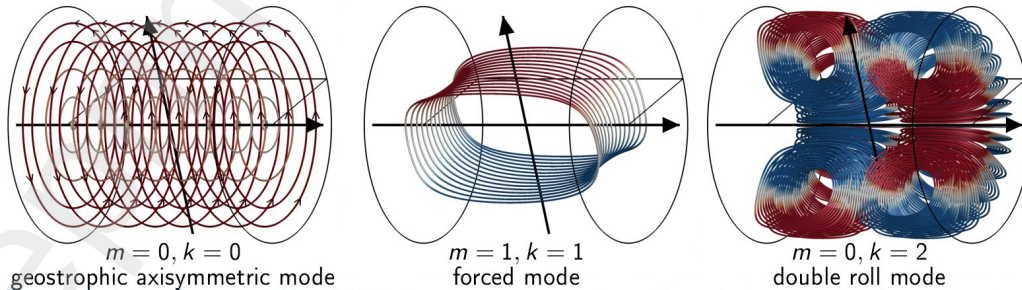


Figure 2: Three main modes in a fluid flow in a precessing cylinder [9].

Characterizing the flow state inside the rotating cylinder is essential for understanding the underlying flow dynamics, validating numerical simulations, and assessing the conditions required for magnetic field generation. However, flow measurements in this experiment pose several challenges. Due to the opacity of liquid sodium, optical measurement techniques such as Particle Image Velocimetry (PIV) or Laser Doppler Anemometry (LDA) cannot be applied [12]. Invasive probes based on mechanical or thermal anemometers would disturb the flow and are difficult to integrate into the sealed sodium container. Pressure sensors only capture information at the outer perimeter of the vessel, which does not provide sufficient information to reconstruct the amplitudes of the dominant inertial modes or otherwise gain insight into the internal flow structure.

Ultrasonic Doppler Velocimetry (UDV), on the other hand, is a proven method for measuring flow velocities in opaque fluids. It has been successfully applied in a variety of applications involving water, suspensions and liquid metals, including sodium [13, 14, 15, 12, 16, 17]. This makes UDV a promising candidate for flow measurements in the DRESDYN precession experiment. However, UDV provides only one-dimensional velocity information along the ultrasound beam and is typically used to investigate local flow phenomena in stationary laboratory experiments. More generally, most flow diagnostics, including PIV and UDV, are designed to characterize local velocity fields or small-scale structures within accessible regions of the flow. In contrast, the DRESDYN precession experiment requires information about the global flow state governed by large-scale inertial modes extending across the entire vessel. This leads to a fundamentally different measurement task compared to conventional laboratory flow diagnostics: the estimation of mode amplitudes through spatially distributed measurements followed by a reconstruction of the global flow state.

In this work we outline the concept and technical foundation of a distributed measurement system for capturing large-scale flow structures in the DRESDYN precession experiment using Ultrasonic Doppler Velocimetry and focus on the implementation and validation of the ultrasound subsystem of the sensor nodes forming the basis of this system. Starting from a system-level description of the future measurement system (Section 2), we derive requirements for the ultrasound subsystem of sensor nodes (Section 3) and compare these against commercially available systems as well as systems developed in a scientific context (Section 4). In Section 5 we present a custom hardware design and describe its building blocks. Validation measurements are presented in Section 6, followed by first-time measurements of radial velocity profiles in a down-scaled model of the final experiment (Section 7).

2. Concept and System Design of Distributed Measurement System

Since the flow in the precession experiment is dominated by large-scale inertial modes extending across the entire vessel, measurements at a single location are not suitable to characterize the global flow state. Instead, the amplitudes of the dominant inertial modes must be inferred from velocity measurements obtained at multiple spatially distributed locations.

Ultrasonic Doppler Velocimetry provides one-dimensional velocity information along the ultrasound beam and therefore captures only a small subset of the flow field at each measurement location. However, by combining measurements from multiple sensors placed at carefully selected positions, the measured velocity components can be made sensitive to specific inertial modes. Numerical simulations indicate that the global flow state can be approximated by a small number of dominant low-order inertial modes. Consequently, estimating the amplitudes of these modes from a set of localized velocity measurements enables an approximate reconstruction of the large-scale flow structure.

In the experimental setup, sensor access to the vessel's interior is provided through a set of predefined flanges located on the side wall of the cylindrical container. These flanges determine the feasible sensor and measurement positions that can be used in the distributed measurement approach. By selecting a subset of these locations that provide access to measurement locations with velocity components sensitive to the dominant inertial modes, the distributed measurement concept can be implemented within the geometric constraints of the experiment.

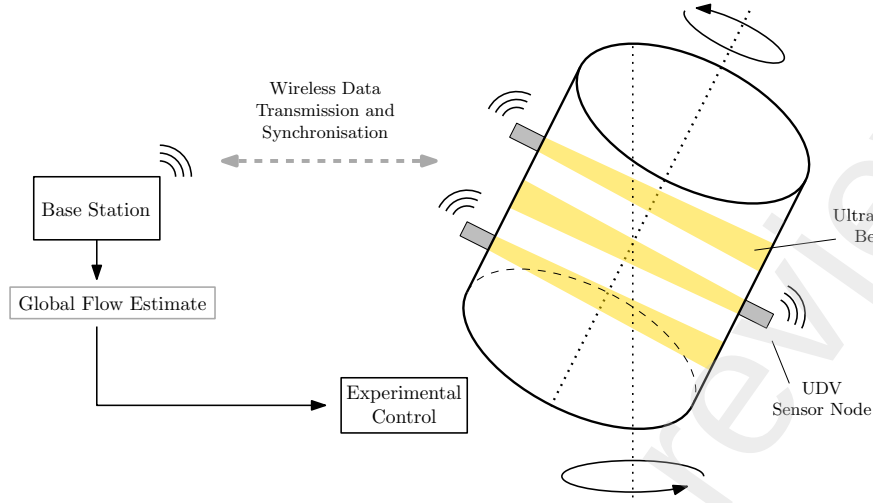


Figure 3: High-level system overview of the planned distributed measurement system. Multiple sensor nodes are mounted to the double-rotated vessel, take localized UDV measurements and transmit them wirelessly to a base station. The estimated global flow state can be used by the experimental control to reach dynamo action.

As a technical basis, we propose a system consisting of multiple sensor nodes integrated into a subset of the available flanges and an external base station. Due to integration requirements of the experiment, the sensor nodes operate autonomously and are powered locally. Communication and synchronization between the sensor nodes and the base station is realized via a wireless network.

Each sensor node performs synchronized UDV measurements and locally processes the acquired signals to extract relevant flow information and reduce the data rate prior to transmission. The base station combines the information from all sensor nodes to estimate the global flow state inside the vessel and can adapt measurement parameters for individual nodes if required. The information on the flow states can be used to control the experiment to reach dynamo action. Since real-time processing on resource-constrained sensor nodes inevitably involves a loss of information, the nodes are additionally capable of storing raw measurement data locally for subsequent offline analysis.

Central aspects of the measurement strategy, such as the optimal placement of sensors and the reconstruction methods used to estimate the global flow state, require further investigation based on flow simulations and experimental studies. To enable these experiments, however, a measurement system providing the required core functionality is necessary. Consequently, this work focuses on the development and validation of extensible sensor nodes providing UDV measurement capabilities, local data storage, and ba-

sic wireless communication, as well as their integration into the experimental setup.

3. Hardware requirements of UDV subsystem

The UDV subsystem has to be capable of performing measurements within a parameter range suitable for the flow conditions in the DRESDYN precession experiment as well as the down-scaled model used for preparatory experiments (see section 7). From the signals captured by these measurements, velocity profiles can be estimated. Raw signals need to be stored persistently on the device and computational resources for on-device processing need to be provided. Finally, an integration into the experiment in terms of size and wireless operation is required.

3.1. UDV parameters

A general description of the measurement principle of UDV can be summarized as follows [13]: An ultrasonic burst consisting of a series of pulses with a frequency f_e is emitted from a transducer. The transducer receives echoes scattered from the surfaces of the particles within the fluid. Based on the speed of sound c within the liquid, the position of these particles within the measurement depth d can be calculated from the delay between emission and reception. By evaluating consecutive emissions with repetition rate f_{PR} the speed of these particles along the insonation axis can be calculated.

However, the measurement principle in its basic form imposes limitations on the detectability of combined large flow velocities and large measurement depths [18, p. 178].

The measurement depth d limits the maximum pulse repetition frequency f_{PR} , since a pulse can only be sent once the last relevant echos from the previous emission have arrived:

$$f_{PR} \leq \frac{c}{2d} \quad (1)$$

If phase shift between echos from the same particle arrangement from subsequent emissions exceed $\pm\pi$ an incorrect lower flow velocity is detected (Aliasing). This limits the maximum detectable flow velocity v_{max} for a given transmission frequency f_e :

$$v_{max} = \frac{cf_{PR}}{4f_e}. \quad (2)$$

From 1 and 2 we can conclude that the emission frequency is limited by the measurement depth and maximum expected velocity inside the experiment:

$$f_e = \frac{cf_{PR}}{4v_{max}} \leq \frac{c^2}{8dv_{max}} \quad (3)$$

Table 1 summarizes the relevant experimental conditions and derived limitations of measurement parameters for the large-scale as well as the down-scaled preparatory experiment detailed in Section 7.

Parameter	Down-Scaled Experiment	Full-Scale Experiment
Medium	Water	Liquid Sodium
Temperature of the medium	20 °C	120 °C
Speed of sound c	1480 m/s	2500 m/s
Max. measurement depth $d_{\max}$	0.16 m	1.00 m
Max. expected radial velocity $v_{r,\max}$	1.94 m/s	2.36 m/s
Max. pulse repetition rate f_{PR}	4625 Hz	1250 Hz
Max. emission frequency f_e	882 kHz	331 kHz

Table 1: Estimated experimental conditions and derived limitations for measurement parameters for the down-scaled and full-scale experiment. Estimated radial velocities are extrapolated from simulation data at lower rotation speeds. The maximum measurement depths are defined by the radius of the respective vessel.

In order to capture the maximum expected velocities over the entire measurement depth, the transmission frequencies would have to be comparatively low.

This would have several undesired effects on the measurement. For one, an increase in wavelength reduces axial resolution, however this is less problematic when capturing large-scale flows structures. A more severe limitation is the widening of the ultrasonic beam defined by the beam divergence angle β . It can be approximated from the speed of sound c , the emission frequency f_e and the diameter of the transducer D according to the following formula [19, p. 71]:

$$\beta = \arcsin\left(\frac{1.22c}{Df_e}\right). \quad (4)$$

The limited integration space in the sodium experiment only allows for transducers with a maximum diameter $D_{\max} = 12$ mm. Together with the maximum emission frequency determined in Table 1 this would result in a wide -6 dB beam divergence angle of $\beta = 50.2^\circ$ for the sodium experiment. As a result, the larger area covered by the beam for deep measurements reduces lateral resolution and signal-to-noise ratio, since only a small fraction of the emitted energy is scattered back to the transducer. Furthermore, at such a large diversion angle the beam turns into a cone leading to geometric measurement errors such as sensitivity towards non-axial velocity components.

At this point it is clear that the transmission frequency will have to be selected dynamically for a certain measurement depth as the result of a tradeoff between maximum flow velocity, axial and lateral resolution and signal-to-noise ratio (SNR). While the values in Table 1 would serve as a lower bound, we assume $f_{e,\max} = 2$ MHz (with a resulting divergence angle $\beta_{e,\max} = 7.3^\circ$) as an upper bound.

For the hardware design of the UDV subsystem only the maximum possible transmission frequency is relevant since it determines the required maximum sampling rate and maximum data throughput. Therefore we conclude that a maximum sampling rate of $f_s = 16$ MHz will be sufficient to support all conceivable measurement scenarios.

3.2. Persistent storage

The measurement system needs to store all raw ultrasound data captured during an experimental run, in order to enable subsequent fine analysis. To allow for experimentation with arbitrary transmit-receive patterns the system must sustain an average throughput of the full sampling data rate of the analog-to-digital converter (ADC) into a persistent storage. The required size of the storage is defined by the expected maximum length of an experimental run of 4 h. Since physical access to the experiment is restricted it is furthermore important that the acquired data can be read out quickly after an experimental run to not delay the experimental schedule.

3.3. Integration

Due to the high rotation and precession rates, the experiment is subject to extreme mechanical stresses [20]. This imposes strict constructive restrictions on the integration of any measurement system into the experiment. The sensor flanges are welded into the cylindrical container and are specifically designed to withstand all experimental conditions. These flanges provide a circular mounting platform with a predefined bolt circle limiting the available area of a measurement system to a width of 45 mm and a length of 120 mm. In order to minimize centrifugal forces and meet clearance requirements, a mounted measurement system must not be higher than 100 mm.

The surface of the ultrasonic transducer must be shielded from the chemically aggressive sodium by a protective housing. In prior experiments at HZDR utilizing an experimental sodium loop, housings made from stainless steel enabled UDV measurements after having undergone further mechanical, thermal, and chemical treatments to ensure constant coupling into the medium [14].

The sensor nodes will be operated in a double-rotated system, making it challenging to connect them to a wired power supply and control system. To

ease integration a battery based powering scheme and wireless control system is required. The battery must sustain operation for at least the duration of an experimental run and allows for quick battery replacement during short time windows between runs.

3.4. Processing

To enable experimentation with on-device processing, the system needs to provide computational resources in terms of processing power and volatile storage. While the required processing power strongly depends on the desired measurement rate and the evaluation algorithm used, the required amount of volatile storage can be approximated from the measurement parameters. UDV measurements consisting of $N = 100$ echoes have been proven to deliver accurate velocity estimations [21]. Providing volatile memory for two measurements capturing the full measurement depth $d_{\max}$ can be considered sufficient to store both the captured signals of one measurement and its intermediate processing results. This leads to a required amount of volatile memory M_{vol} defined in 5 or $M_{\text{vol}} = 1.28 \text{ MS}$ with the parameters for the sodium experiment from Table 1.

$$M_{\text{vol}} = \frac{2Nd_{\max}f_s}{c} [\text{samples}] \quad (5)$$

4. State of technology - UDV measurement systems

Complete systems for distributed UDV measurements are currently not available. However, there are several commercial and scientific measurement systems with demonstrated UDV capabilities in various applications. Their suitability and adaptability as the UDV subsystem of a sensor node will be examined below.

Both the *DOP* series manufactured by *Signal Processing SA* [22] and the *us4R* devices from *us4us sp. z o. o.* [23] are capable of performing multichannel measurements with the parameter space outlined in Section 3.1. Both systems provide capabilities for on-device processing but are designed around a host PC connection for data storage and do not support standalone operation. Together with the power supply requirements and their form factor, this renders integration into the experiment impossible.

The *Peacock UVP* from *Ubertone* [24] meets the form factor requirements for integration into the experiment. However, the device calculates flow velocities within predefined volumes directly from received ultrasound echoes and transfers those to a host system. Storage of raw ultrasound data for reprocessing with adjusted evaluation parameters is not possible.

The open source hardware platform *unOrick* [25] comprises a single-channel ultrasonic measurement system that can be freely configured using a field-programmable gate array (FPGA). Due to the availability of design files, adjustments to fulfill the form factor requirements are possible. The platform is designed for use with a host PC or single-board computer and does not provide for local raw data storage or sufficient volatile memory for on-device processing.

In the field of medical technology, the *Wulpus* [26] and *TinyProbe* [27] platforms represent approaches for significant miniaturization of ultrasound systems. In line with their area of application, these systems focus on multi-channel transducers and non-continuous measurement scenarios. As open-source platforms, both systems would allow for adaptation to the experimental requirements. However, their core components impose fundamental limitations. The microcontroller (MSP430FR5043, Texas Instruments) used in *Wulpus* does not have sufficient internal memory or interfaces for external memory with the required bandwidth. Neither on-device processing nor raw data storage at the required rates is feasible. *TinyProbe* utilizes a microcontroller (STM32F411VEH6, ST Microelectronics) with interfaces for persistent storage of the required capacity but lacks sufficient volatile memory or interfaces to expand it. Furthermore, the bandwidth of the interface to the FPGA providing the data from ultrasound measurements is limited to 25 Mbit/s.

	Acquisition Parameters	Raw Data Storage	Processing Capabilities	Form Factor	Wireless Control	Battery Power
DOP	✓	✗	✓	✗	✗	✗
us4r	✓	✗	✓	✗	✗	✗
Peacock UVP	✓	✗	✓	✓	✗	✗
unOrick	✓	✗	✗	✓	✗	✗
Wulpus	✓	✗	✗	✓	✓	✓
TinyProbe	✓	✗	✗	✓	✓	✓

Table 2: Overview of existing UDV measurement systems and their suitability for the application

The systems capable of calculating velocity profiles on the device do not meet the requirements of the experiment in terms of the form factor, wireless

control or battery power necessary for integration. As these are proprietary systems, adaptation to the experimental requirements is not feasible. In contrast, significant miniaturization of ultrasound measurement systems is being demonstrated in the field of medical technology in particular. Given their application, they are in line with the integration requirements of the experiment in terms of wireless control and battery power. However, this also results in system designs with sample data transfer bandwidths that are insufficient for the intended application. There is currently no system that combines miniaturization, on-device processing, and local storage of raw ultrasound data. The development of a UDV measurement system meeting these requirements is therefore necessary to form the basis of a sensor node for the distributed measurement system.

5. IUP - Integrated Ultrasound Platform

The distributed measurement system outlined in Section 2 requires a sensor node capable of executing UDV measurements and persistently storing the unprocessed measurement data for further analysis. Through a wireless interface measurements can be started and stopped and status information can be transferred to a remote device. An integrated ultrasound platform providing this functionality has been developed. The modular design is constructed along the main functional units described in the following. An overview of its building blocks is depicted in Fig. 4.

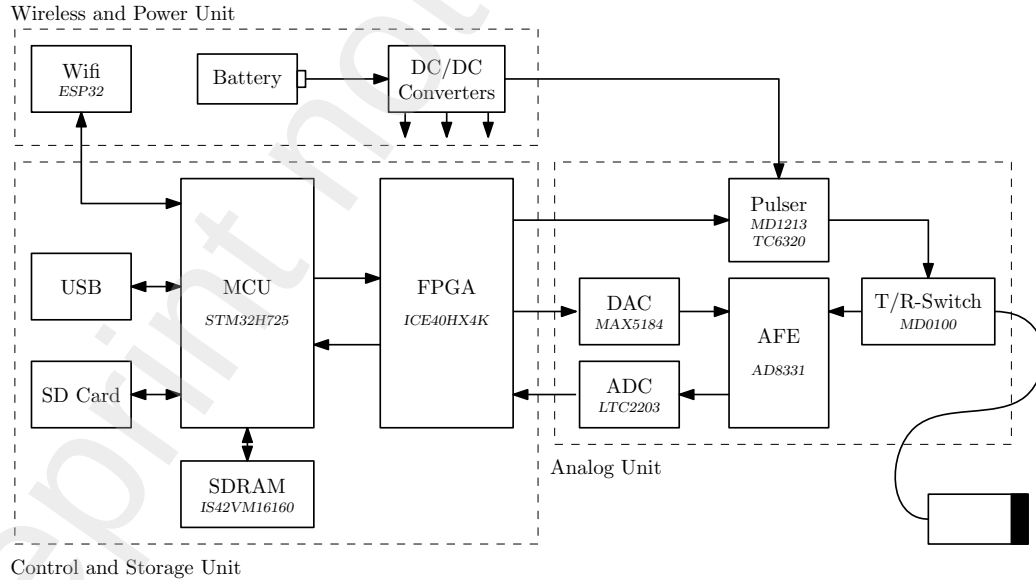


Figure 4: Overview of IUP building blocks and distribution onto functional units

5.1. Analog Unit

This unit is responsible for generating high-voltage pulses to drive an ultrasonic transducer as well as amplifying and digitizing received echo signals. In order to provide this functionality within minimal printed circuit board (PCB) area the circuitry utilizes integrated circuits commonly used in ultrasound applications. A 3-state pulser circuit (*MD1213* and *TC6320*, *Microchip Technology Inc.*) controlled by an FPGA generates bipolar pulses at the selected emission frequency f_e . A symmetric pulse voltage of ± 32 V is generated by a dual DC-DC converter (*LT8582*, *Analog Devices, Inc.*). The electrical signals generated by the transducer from received echoes are amplified within an analog front-end (AFE, *AD8331*, *Analog Devices, Inc.*). A transmit-and-receive (T/R) switch protects the AFE circuitry from high-voltage pulses (*MD0100*, *Microchip Technology Inc.*). In order to compensate for damping within the fluid, time gain compensation is applied by varying the gain of a voltage-controlled amplifier within the AFE. The control voltage is generated by a digital-to-analog converter (DAC, *MAX5184*, *Analog Devices, Inc.*) controlled by the FPGA via a parallel interface. Following amplification, the signal is processed by an analog anti-aliasing filter and digitized by an ADC (*LTC2203*, *Analog Devices, Inc.*). An ADC resolution of 16 bit was chosen in order to maximize dynamic range for subsequent raw data analysis. Sample data is transferred from the ADC to the FPGA via a parallel interface.

5.2. Control and Storage Unit

The control and storage unit at the center of the system is responsible for executing the Doppler measurements within their timing constraints, storing the acquired data on persistent storage and exposing interfaces for remote control. Its architecture is centered around a microcontroller unit (MCU, *STM32H725IGK*, *ST Microelectronics NV*) and a field-programmable gate array (*ICE40HX4K*, *Lattice Semiconductor Corp*). This architecture and the selected parts have multiple advantages for the targeted application. FPGAs allow for cycle-accurate measurement control through their deterministic behavior and provide infrastructure for implementing flexible parallel high-speed interfaces for sample data transfer. The *STM32H7*-series, on the other hand, provides peripherals for interfacing with high bandwidth persistent memory as well as parallel interfaces for data transfer with the FPGA. Furthermore its core running at up 480 MHz provides strong signal processing capabilities.

The FPGA controls the analog circuitry and receives samples from the ADC. With an ADC data width of $B = 16$ bit and a sampling frequency of

$f_s = 16$ MHz, the maximum average data rate $R_{\max}$ required for the following path to persistent storage is defined by:

$$R_{\max} = f_s \cdot B = 32 \frac{\text{MB}}{\text{s}} \quad (6)$$

After an internal first-in first-out (FIFO) buffer, which accounts for bursty operation of downstream interfaces, the sample data is forwarded through a parallel interface with a capacity of 64 MB/s to the microcontroller unit (MCU). There it is buffered on external memory (SDRAM, *IS42VM16160K-6BLI, Integrated Silicon Solution*) before being written to an SD card for persistent storage. The SDRAM operates at 100 MT/s on a 16 bit bus and is interfaced by a memory controller inside the MCU. Read and write access to the SDRAM is realized via direct memory access controllers (DMA) freeing up the core of the MCU for control tasks. DMA is also responsible for transferring sample data from SDRAM to persistent storage. For continuous measurements within an experimental run with an approximated duration of up to 4 h and data rate $R_{\max}$ at least 460.8 GB of persistent sample data storage are required. With a capacity of up to 2 TB (SDXC standard, [28]) and a minimum average sequential write speed of 30 MB/s (UHS Speed Class 3, [29]) SD cards match the bandwidth and capacity requirements, while also allowing for easy replacement and data transfer to a PC.

A detailed overview of the sample data flow and the data rates of the involved interfaces is depicted in Fig. 5. In practice, a duty cycle of 55%¹ showed stable performance during evaluation measurements. This is due to two DMA units accessing the Memory Controller concurrently without adjustable prioritization leading to overruns in the FPGA FIFO before the SDRAM buffer is full.

5.3. Wireless Interface and Power Unit

Due to the constraints imposed by the integration into the experiment, wireless control as well as a local power source are required. The wireless interface for measurement control is realized with a Wi-Fi module (*ESP32-S3-MINI-1U, Espressif Systems*). Control commands are received via Wi-Fi and forwarded to the MCU via an asynchronous serial interface. Status messages from the MCU are transmitted in the reverse direction. Lithium-ion batteries are chosen as power sources because of their superior energy density and availability in standardized form factors. Batteries in the 18650 form factor fulfill the constraints in terms of available integration volume

¹100 x 13312 samples at 16 MS/s during a measurement cycle of 150 ms

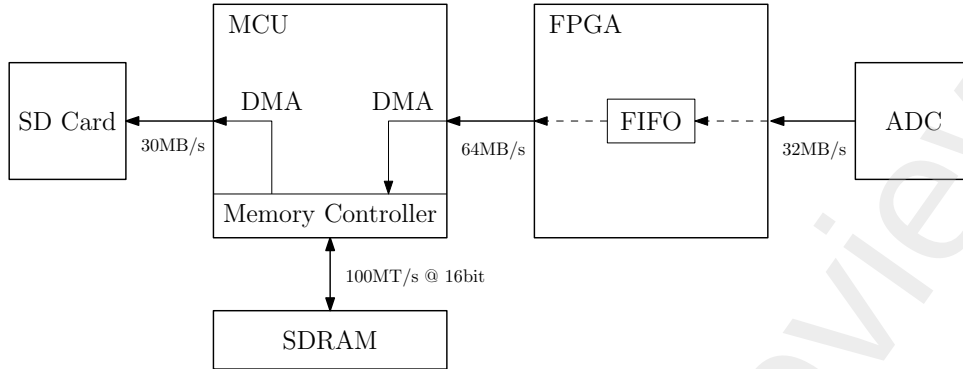


Figure 5: Flow of sample data inside IUP with maximum average throughput of interfaces.

and provide up to 14.4 W h. As battery voltage depends on the state of charge and load current, a DC-DC converter generates a stable voltage of 5.5 V, providing sufficient headroom for subsequent linear regulators. These then power the different system units with their respective operating voltages (Analog: 3.3 V, 5.0 V, Digital: 1.2 V, 1.8 V, 3.3 V). During normal operation, including continuous measurement, storing data on SD card and status logging via Wi-Fi, the measured power consumption of IUP is 2.96 W. This allows for sustained data acquisition throughout a full experimental run in the DRES-DYN precession experiment with the charge provided by a single 18650 cell.

5.4. Integration

The IUP is designed following a modular approach as a stack of three PCBs (see Fig. 6). Each PCB contains the circuitry of one functional unit as described above. This allows for exchanging PCBs of functional units as development of the sensor node progresses. The compact size of the assembled stack (62 mm x 45 mm x 23 mm) and its low weight (43.4 g) allow integration of the complete measurement system into the sensor node considering the constraints described in Section 3.3. A custom sensor node housing, depicted in 7, ensures mechanical integrity of the PCB stack and constant transducer coupling during experimental runs. The transducer is installed in a cylindrical adapter made from stainless steel. To ensure coupling of ultrasound energy into the liquid sodium, the adapter is manufactured following the design principles and mechanical and chemical treatments outlined in [14]. A thermally stable grease (*Super Adhesive Grease*, *CRC Industries Inc.*) is applied between transducer and adapter to ensure constant acoustic coupling. The full functionality of the sensor node under mechanical stress, including the battery, interconnects and transducer coupling, has been vali-

dated in a centrifuge. An external 2.4 GHz antenna is placed on the outside of the sensor node housing to ensure optimal connectivity.

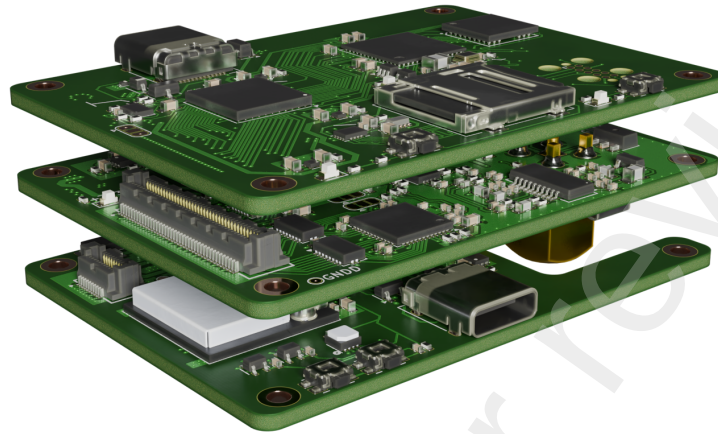


Figure 6: IUP PCB stack with Control and Storage Unit, Analog Unit and Wireless Interface and Power Unit (from top to bottom)

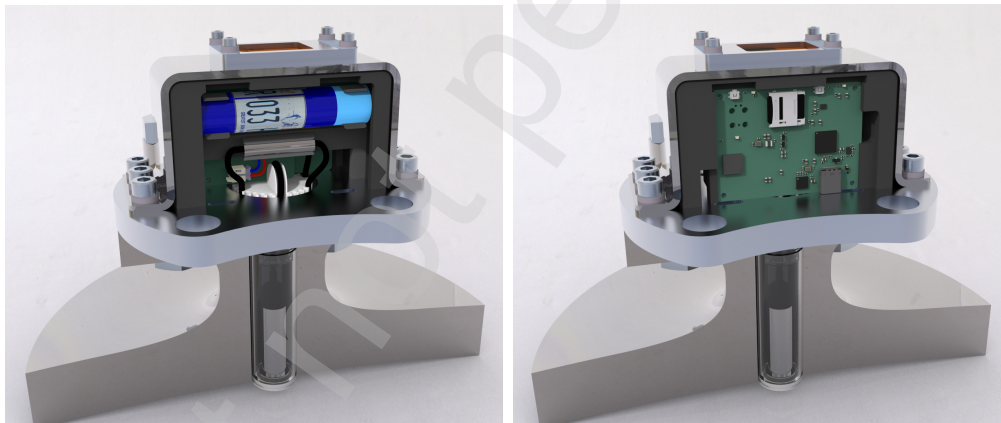


Figure 7: Integration of IUP and Transducer into a DRESHDYN sensor flange

6. Validation

In order to validate the functionality of the IUP measurement system and study its usability for the intended application, a comparative flow velocity measurement against a PIV system was conducted. Therefore, an open water channel with a cross section of $31\text{ cm} \times 41\text{ cm}$ and a length of 5 m [30] was used. It was filled with water seeded with white particles with an average

diameter of $100\text{ }\mu\text{m}$. By varying the power of the circulation pump, different static flow states were established and simultaneous measurements were taken with both systems. All raw data captured during this experiment is available in [31].

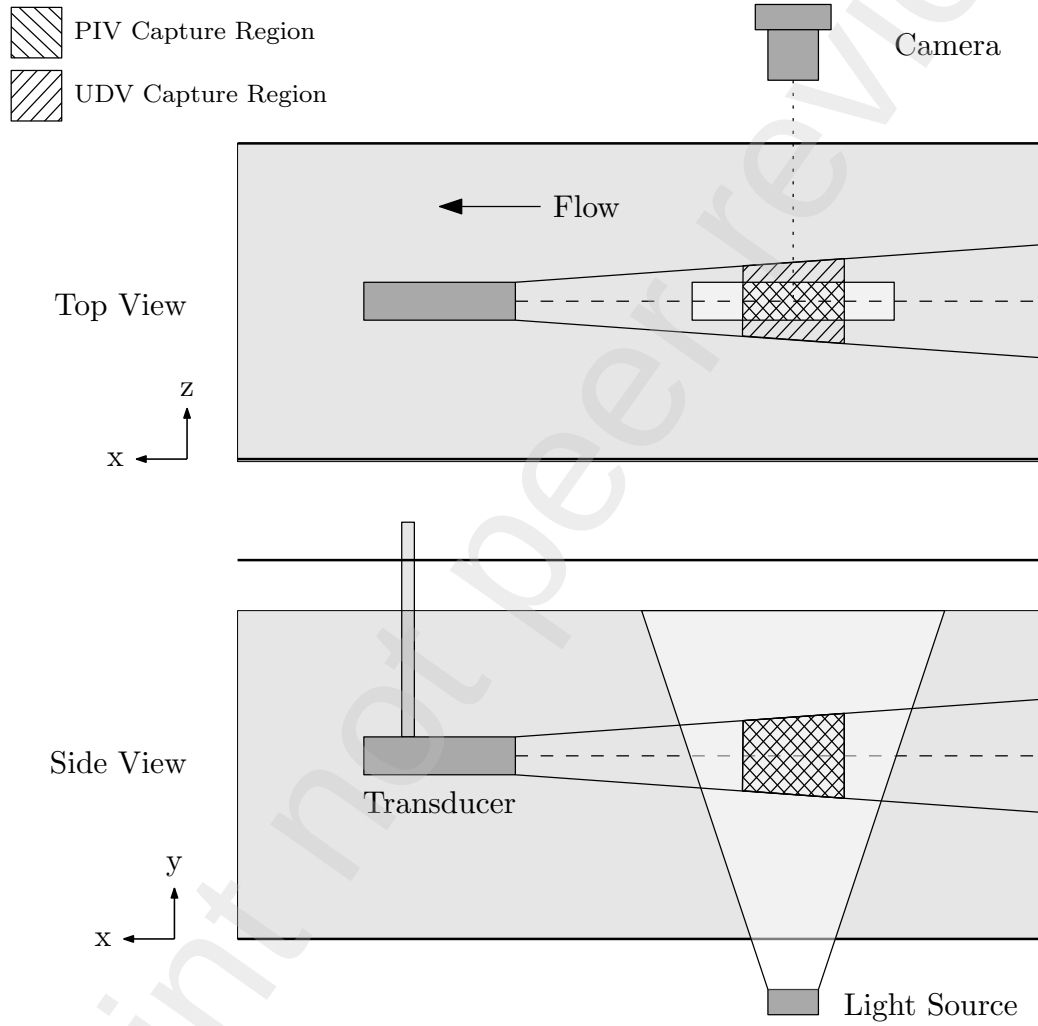


Figure 8: Combined PIV and UDV measurement setup including coordinate system and capture regions (not to scale)

Fig. 8 gives an overview of the measurement setup: A transducer (*TR0210LH*, *Signal Processing S.A.*) was mounted in the center of the channel pointing upstream and connected to the IUP. The PIV system consisted of a constant light source (blue laser diode, *PLPT9 450LB_E*, *ams-OSRAM AG*) that illuminated the channel through a window in the floor and a high speed camera

(*Phantom VEO 710L*, *Vision Research Inc.*) capturing images through a transparent side wall. Calibration was conducted by placing a conventional ruler in the center of the light sheet. Raw ultrasound data was recorded to the SD card of the IUP and processed after the measurement on a PC with custom software tools while *DaVis* (*LaVision GmbH*) was used to process the PIV images. Operation and processing parameters were selected as outlined in Table 3.

UDV Parameters

Emission	4 periods at $f_e = 2$ MHz
Emissions per Profile	100 with $f_{PR} = 727$ Hz
Pulse Amplitude	46 V _{pp}
AFE Gain	30 dB
Sample Rate	16 MHz
Bandpass	200 kHz bandwidth 10 th order butterworth
Doppler Algorithm	2D Autocorrelation [32]
Speed of Sound	1465 m/s (15 °C)
Range Gates	128 of 104 samples (4.8 mm width)
Duration	100 profiles at 150ms spacing

PIV Parameters

Frame Rate	40 Hz
Image Size	1280 × 800 px
Window Size	32 × 32 px with 50 % overlap
Scaling Factor	3112.84 px/m
Duration	900 images

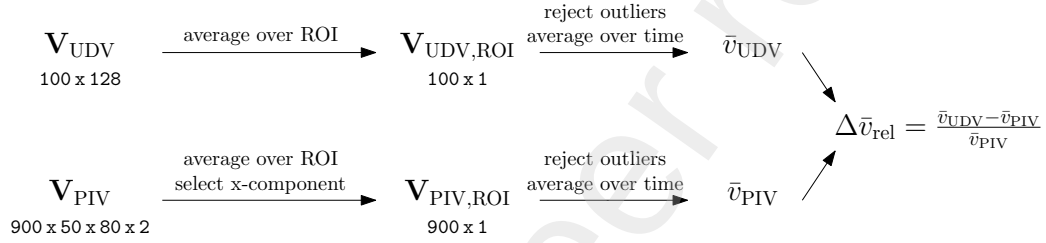
Table 3: UDV and PIV operational and processing parameters

Due to its measurement principle, UDV can only capture axial velocity components (along the x -axis). Furthermore, sound energy is emitted approximately as a cone-shaped beam that widens with distance so all UDV measurements must be interpreted as mean x -velocities over the beam cross-section in the yz -plane. PIV on the other hand captures a field of xy -velocities over the xy plane - averaged over the extension of the light sheet in z -direction.

To compare those different types of measurement data, a three-dimensional region of interest was defined whose mean x -velocity component can be captured by both systems. That region was chosen as a set of consecutive range gates of the ultrasonic velocity profile located in the center of the PIV image.

By selecting a subregion of the PIV vector field that matches the depth range and the shape of the ultrasonic beam in the xy -plane, the same volume is captured - except for the extension in z -direction (see Fig. 8 for illustration). For UDV, this dimension is determined by the shape of the ultrasonic beam at the region of interest (average -6 dB width of 15 mm), for PIV by the thickness of the light sheet ($1/e^2$ width of 30 mm). Since both values are similar and small compared to the width of the channel this mismatch is assumed to be insignificant.

Figure 9: Data dimensionality and processing steps to compare a set of UDV and PIV measurements for a single flow state



Once a flow state was set, a series of measurements ($\mathbf{V}_{\text{UDV}}$ and $\mathbf{V}_{\text{PIV}}$) were taken with both systems and averaged over the region of interest. While for UDV this represents averaging over a set of range gates, the PIV velocity vector field had to be upsampled before averaging the x-components of all vectors located inside the region. The result is a timeseries of mean x-velocities inside the systems' capture regions ($\mathbf{V}_{\text{UDV,ROI}}$ and $\mathbf{V}_{\text{PIV,ROI}}$). After removing outliers - defined as values with a distance of more than three standard deviations from the mean of the series - a mean x-velocity during the measurement duration can be calculated for each system ($\bar{v}_{\text{UDV}}$ and $\bar{v}_{\text{PIV}}$). Those values together with the estimated probability density functions are shown in Fig. 10. The distributions have a similar shape and their moments, i.e. mean and standard deviation, are comparable. Since the widths of the distributions are similar for UDV and PIV, this suggests that they are determined primarily by the measurand, e.g. turbulence, rather than by the measurement itself.

For direct comparison we declare PIV as a reference and define $\Delta\bar{v}_{\text{rel}}$ as the relative deviation of UDV to PIV measurements for a single setpoint. Fig. 11 shows $\Delta\bar{v}_{\text{rel}}$ for all setpoints. The result suggests a systematic error of -2.2% independent of setpoint (and therefore flow velocity). This result should be interpreted as a comparison between two independent measurement techniques rather than as a direct estimate of the absolute accuracy of the UDV system. Several experimental factors may contribute to this de-

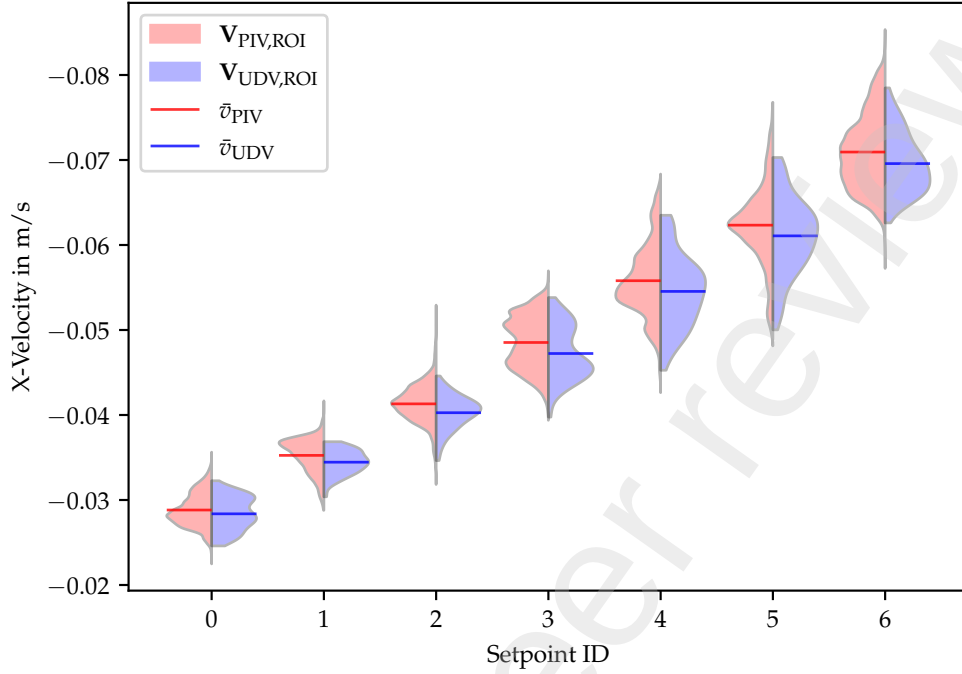


Figure 10: Estimated probability density functions of mean x-velocities over respective capture regions for both systems and multiple setpoints after outlier rejection. Solid lines indicate arithmetic means of the distributions. Estimates are based on gaussian kernels with a bandwidth selected according to Scott's Rule [33, p. 131].

viation, including the assumed sound speed in water, slight transducer misalignment, the finite measurement volume of the ultrasound beam, as well as uncertainties related to PIV calibration, optical mapping, and interrogation procedures. In addition, small geometric deviations of the experimental setup and differences in spatial and temporal averaging between the two techniques may influence the comparison. Under the present experimental conditions, the individual contributions of these effects cannot be separated. A rigorous determination of the absolute accuracy would require a dedicated reference experiment and a comprehensive uncertainty analysis of both measurement techniques, which was beyond the scope of this study. Regarding the DRESYN experiment however a small and constant systematic error is not problematic for capturing large scale flow structures or relative amplitudes of modes.

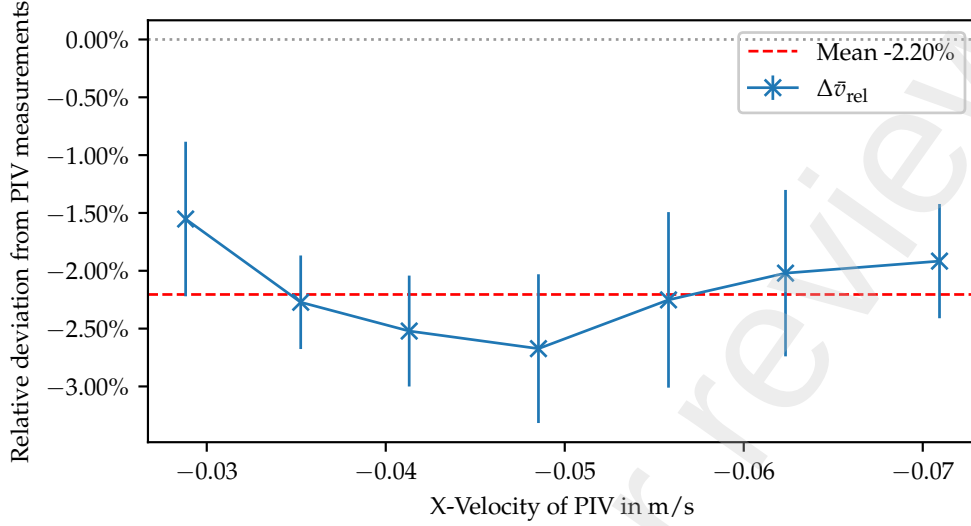


Figure 11: Relative deviation of UDV measurements compared to PIV for a set of flow states. The errorbars represent a single standard deviation and are calculated through uncertainty propagation from $\bar{v}_{UDV}$ and $\bar{v}_{PIV}$. The mean employs weighting by inverse variance.

7. Radial measurement on Down-scaled Experiment

In preparation of the precession experiment with liquid sodium a down-scaled experiment (scale 1:6) filled with water is in operation [34]. The cylinder with radius $R = 163$ mm is driven by an asynchronous motor and can reach rotational frequencies of up to 10 Hz. The container in turn is mounted on a turntable driven by a second asynchronous motor and can achieve frequencies of up to 1 Hz. It has already been used to validate flow field simulations with flow velocity measurements utilizing an ultrasound doppler velocimeter (*DOP3010, Signal Processing S.A.*)[34, 6].

A sketch of the setup is shown in Fig. 12. One of the poles of the vessel embeds a set of ultrasonic transducers at different radial locations. Through a multiplexer and two sliprings, one of them at a time can be routed to the outside of the experiment for capturing axial flow velocities inside the double rotated vessel. However, this design does not allow measuring radial velocity components as required in the large scale experiment.

Since the IUP is battery powered and controlled wirelessly it is not bound to these constructive limitations and thus enables first-time capturing of radial velocity components in the preparatory experiment. The device and a connected transducer (*TR0210LH, Signal Processing*) are mounted to the side wall of the acrylic vessel with a 3D-printed fixture for radial acoustic

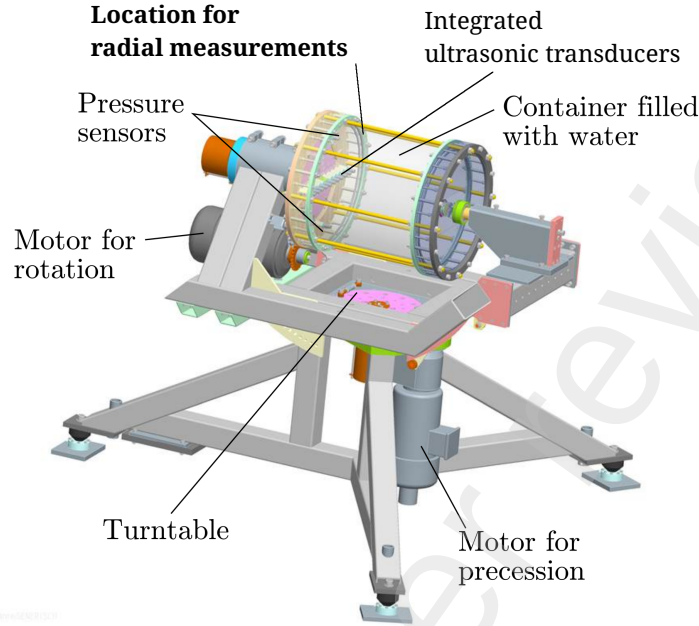


Figure 12: Sketch of the down-scaled experiment filled with water and location of radial measurements (adapted from [34]).

access to the medium. The water was seeded with copolyamid particles (*Griltex 2A P1*, 60 % 50 μm , 40 % 80 μm). For a fixed precession ratio, a set of rotation rates were selected as static setpoints. After the flow had settled in a static state, UDV signals were captured, stored on device and processed offline as described in Section 6.

Fig. 13 shows the radial velocity over time measured for a fixed rotation rate at an axial position matching a possible sensor location in the large-scale experiment (see Fig. 12). Since the precession ratio was set well below the transition point to the turbulent regime, the flow is expected to consist mainly of the modes $(m, k) = (0, 0)$ and $(m, k) = (1, 1)$ (see Fig. 2 for illustration). The tangential flow of the axisymmetric mode is perpendicular to the radial insonation axis and can therefore not be captured in this setup. The forced mode represents a standing wave in the reference frame of the turntable. At the poles of the vessel the direction of flow is mostly radial and can thus be captured by the IUP. Since the vessel is rotating around the standing wave, we observe periodic amplitude modulations with the rotation rate of the vessel.

In the present setup (with one single UDV probe), the Fourier amplitude of the measured radial velocity component can be used directly as an approx-

imate measure of the dominant $(m, k) = (1, 1)$ mode amplitude. However, the extracted Fourier amplitude may contain contributions from all modes with the same temporal or azimuthal signature so that a unique separation of the axial modal structure is only possible if higher axial modes with $k > 1$ are known to be significantly weaker or by combining simultaneous measurements at different locations.

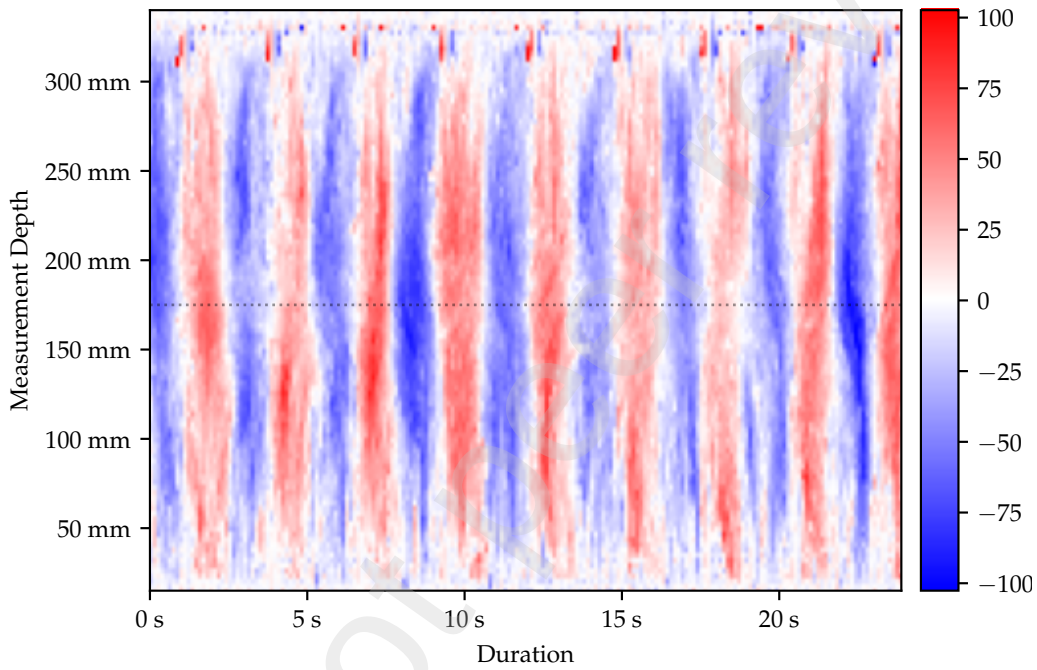


Figure 13: Radial flow velocity over time captured 4.6 cm from a pole at a rotation rate of 0.36 Hz and a precession ratio of 0.076. The dotted line represents the rotation axis of the cylinder.

For a quantitative analysis we extracted the periodicity of the amplitude modulations at a single location in the center of the vessel (as indicated by the dotted line in Fig. 13) and compared it to the rotation rate measured by an optical encoder built into the experiment. Fig. 14 shows the deviation of the measured rotation rate from UDV from to the rotation rate measured by the optical encoder. Given the simple design of the optical encoder and the short measurement duration of only a couple of complete rotations the results show good alignment (deviation of less than 0.6 %) and demonstrate that the IUP is capable to extract high-level flow properties from UDV measurements.

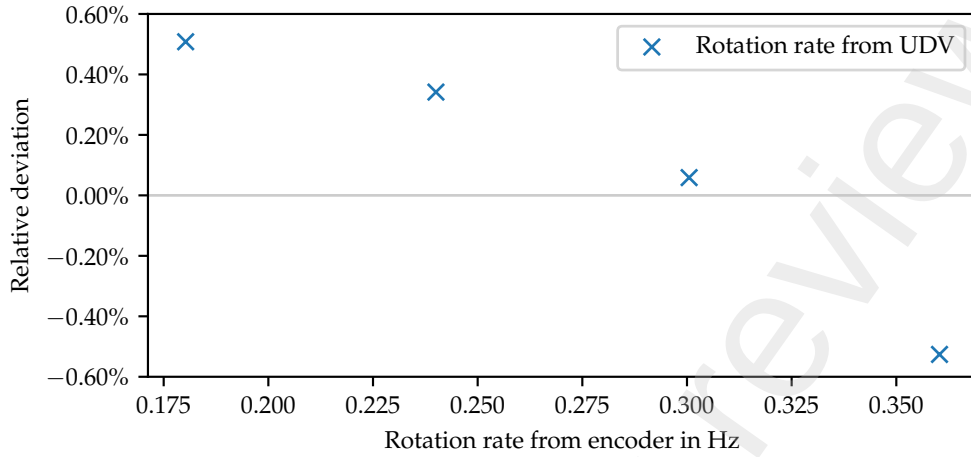


Figure 14: Comparison of measured rotation rate from optical encoder and rotation rate derived from UDV

Going forward, this paves the path for more in-depth studies in the preparatory experiment, such as the simultaneous measurement of axial and radial flow velocities and their validation against simulations as well as evaluating strategies for distributed measurements and flow reconstruction techniques in the large-scale experiment.

8. Conclusions and Outlook

We have outlined a measurement strategy to capture large scale flow structures in the DRES Dyn precession experiment using Ultrasound-Doppler-Velocimetry. It is based on a distributed system of sensor nodes that allows taking localized velocity measurements from inside the rotating vessel. Through a targeted selection of measurements location, these localized measurements enable capturing the amplitudes of inertial modes and - since the flow is primarily described by a set of these modes - enable reconstruction of the global flow state.

From a high-level description of the system we have derived the technical requirements of the UDV subsystem of the sensor nodes. It has to support a range of acquisition parameters, provide a high-throughput data path to persistent storage as well as processing resources and wireless control - all within a small form factor and battery powered.

Since no available systems are able to meet these requirements, we presented the *Integrated Ultrasound Platform (IUP)* and detailed its technical implementation. Validation measurements in an open flow channel showed

good alignment with a Particle-Image-Velocimetry system. Finally we demonstrated first-time capturing of radial velocity profiles in a down-scaled model of the DRESDYN precession experiment. Future work will focus on development of the remaining technical capabilities of the sensor nodes such as on-device processing of UDV data and wireless synchronization, accompanied by flow measurements on the large scale experiment filled with water. With the technical infrastructure in place, a complete distributed measurement system will be designed through a combination of simulation and experimental work.

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